NATURAL LANGUAGE PROCESSING

COEN 541

(Module_1: INTRODUCTION)

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DEFINITION

- The process of making computers to **understand and/or generate** human language.

- The process of computer **analyzing input** provided in a human language, and **converting it** into useful form of representation.

- Field of computer science and engineering that deals with **getting computers make sense of language** using linguistics, statistics, and artificial intelligence.

"Data input/output" of a NLP system can be in the form of text, speech, image, or video.

Primary concern of NLP is getting computers to perform useful and interesting tasks with human languages.

Secondary concern of NLP is helping to a better understand human language.

Components of NLP

- **Basically 2 components.**
  - ✓ Natural Language Understanding
  - ✓ Natural Language Generation
- **Natural Language Understanding**
  - ✓ Mapping of an input into a useful representation.
- **Natural Language Generation**
  - ✓ process of constructing natural language outputs from non-linguistic inputs.
  - ✓ Producing output in the natural language from some internal representation

Both NL understanding & generation are hard tasks.

However, NL Generation is much harder NLP task.

Why NLP is Hard?

- ✓ **Ambiguity: e.g.** How many different interpretations does this sentence have "**I made her duck.**"?
- ✓ **Scale**
- ✓ **Sparsity**
- ✓ **Variation**
- ✓ **Expressivity**
- ✓ **Unmodeled Variables**
- ✓ **Unknown representations**

Corpus

- A collection of text (sentences) or speech files which serves as an input.
  - ✓ Also called a dataset.
  - ✓ Training data needed for a given NLP task.

A type is a distinct word in a corpus.

Vocabulary is the set of types.

|V| is the size of the vocabulary.

Each word in a corpus is a token.

N is the number of tokens in the corpus.

NLP Applications

- **Includes**
  - ✓ Conversational agents (bots)
  - ✓ Information retrieval & extraction (e.g. Text question answering systems)
  - ✓ Machine translation
  - ✓ Opinion and sentiment analysis
  - ✓ Social media analysis
  - ✓ Visual understanding
  - ✓ Essay evaluation
  - ✓ Mining legal, medical, or scholarly literature.

Basic NLP Principles

- **Fundamental concepts to be covered includes;**
  - ✓ Text Processing
  - ✓ Syntax and Semantic analysis: POS tagging, NER & Parsing

- **Text Processing**
  - ✓ steps involved in cleaning, transforming, and preparing corpus for analysis or processing.
  - ✓ Helps prepare text data for further analysis, machine learning, or other NLP tasks.
  - ✓ Basic text processing includes; Tokenization, Normalization (Stop word removal, case folding), Stemming or Lemmatization.

Basic Text Processing

❑ Tokenization

- ✓ breaking down text corpus into smaller chunks of elements called tokens.

❑ Types of Tokenization

- ✓ Word Tokenization: breaking down or segmenting text into individual words.
  - e.g. "This is an example sentence" → ["This", "is", "an", "example", "sentence"]
- ✓ Sentence Tokenization: breaking down text into individual sentences.
  - e.g. "This is Abel. He is a student." → ["This is Abel.", "He is a student."]

Basic Text Processing Cont.

- **Types of Tokenization (Cont.)**
  - ✓ Subword Tokenization: breaking down words into subwords or wordpieces.
    - e.g. `"unbreakable" → ["un", "break", "able"]`
  - ✓ Regular Expression Tokenization: using regular expressions to split text into tokens based on specific patterns.
    - e.g., splitting on punctuation or whitespace.

Basic Text Processing Cont.

- **Stop Words**
  - ✓ words which occur frequently in a corpus and have very little meaning or little usage.
  - ✓ e.g. a, an, the, in.
  - ✓ removed from the corpus for the sake of text normalization.

- **Normalization**
  - ✓ task of putting words in a standard format.
  - ✓ e.g. case folding,
  - ✓ Case folding reduce all letters to lower case.
    - For some NLP applications (information retrieval, MT, information extraction) cases can be helpful.
    - NB: US versus us are important

Basic Text Processing Cont.

❑ Stemming

- ✓ process of finding similarities between words with the same root words.
- ✓ acts on words without knowing the context.
- ✓ Thus, it's faster but doesn't always yield the desired result.
- ✓ E.g. Porter Stemmer, Lancaster Stemmer

❑ Lemmatization

- ✓ process of grouping together different inflected forms of a word, called lemma.
- ✓ Output of lemmatization is a proper word.
- ✓ It uses a lexical vocabulary to derive the root form.
- ✓ Thus it is more time-consuming than stemming, and is most likely to yield accurate results.
- ✓ It uses a lexical database called WordNet.

NLP Algorithms

- **Syntactic Analysis**
  - ✓ Also known as parsing, is the process of analyzing the syntactic structure of a sentence.
  - ✓ Steps include: Tokenization, POS tagging, NER, and dependency and constituency Parsing.

- **Part-of-Speech (POS) Tagging**
  - ✓ Here, words in the text are assigned a label with corresponding parts of speech components such as nouns, verbs, adjectives, or adverbs.
  - ✓ Takes features like the previous word, next word, and capitalization of the first word into consideration when assigning a POS tag to a word.

NLP Algorithms Cont.

- **Chunking**
  - ✓ Known as shallow parsing.
  - ✓ picking up individual pieces of information and grouping them into bigger pieces.
  - ✓ Chunking is a process that requires POS tagged input, and it provides chunks of phrases as output.
  - ✓ Similar to POS tags, there is a standard set of chunk tags like Noun Phrase (NP), Verb Phrase (VP), etc.

NLP Algorithms Cont.

- **Named Entity Recognition (NER)**

Assignment 1

- Define the following knowledge of language:
  - ✓ Phonology, Morphology, Syntax, Semantics, Discourse & World Knowledge?
- Write a Python program to implement an n-gram based tokenization on a corpus?
- Define Parsing?

谢谢

Thank you